GUIDE 04

MySQL Server and MySQL Workbench Setup

UniMart Monolithic Full-Stack Lab Series

B.Sc. (Hons) IT - Level 2 | Hands-on Laboratory Practical

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Learning outcomes

- Install MySQL Server and Workbench for local development.
- Create a least-privilege UniMart database account and schema.
- Verify connectivity and use Workbench for modelling, queries and inspection.
- Apply safe backup and classroom reset practices.

Prerequisites

- Administrator rights for software installation.
- The ER design from Guide 01.

Suggested student time: 1.5-2 hours

Compatibility note: MySQL Server 8.4 is the LTS database baseline. MySQL Workbench remains in the 8.0 release line and may connect to 8.4, but Oracle notes that some Workbench features may not function with newer server versions. Test the exact classroom image before the lab.

1. Windows installation using MySQL Installer

1. Download the Community MySQL Installer from the official MySQL site.
2. Choose a Custom or Developer Default setup. Select MySQL Server 8.4 LTS where offered, MySQL Workbench, MySQL Shell and required connectors.
3. Use Development Computer configuration. Keep TCP/IP enabled on port 3306 for local use.
4. Choose strong password authentication and set a unique root password. Store it in a password manager, not a text file or screenshot.
5. Create the Windows service and start it automatically.
6. Complete the installation, open Workbench, and test the Local instance connection.

2. macOS and Linux notes

- Use the official MySQL packages for the operating system. Avoid mixing package-manager and manually downloaded installations.
- Install Workbench from the official distribution where supported. On Linux, install prerequisite libraries indicated by the package manager.
- Confirm the server service is running and that localhost:3306 is reachable only as intended.
- For a uniform class environment, Docker Desktop can be used for MySQL Server while Workbench runs natively; the deployment guide uses containers on the VPS.

3. Create the application schema and least-privilege account

```sql
[SQL]
-- Run as an administrative account in Workbench.
CREATE DATABASE IF NOT EXISTS unimart
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_0900_ai_ci;

CREATE USER IF NOT EXISTS 'unimart_app'@'localhost'
  IDENTIFIED BY 'replace-with-a-strong-local-password';

GRANT SELECT, INSERT, UPDATE, DELETE,
      CREATE, ALTER, INDEX, REFERENCES
ON unimart.* TO 'unimart_app'@'localhost';

FLUSH PRIVILEGES;
SHOW GRANTS FOR 'unimart_app'@'localhost';
```

Least privilege: Do not configure the Spring Boot application to use root. The application account should have only the permissions required for its environment. Production migration and runtime accounts may be separated further.

4. Create and test a Workbench connection

7. On the Workbench home screen select the plus icon beside MySQL Connections.
8. Connection name: UniMart Local; hostname: 127.0.0.1; port: 3306; username: unimart_app.
9. Use Store in Vault/Keychain only on a trusted personal computer. Shared labs should not store passwords.
10. Select Test Connection, then open the connection and run SELECT VERSION(); and SHOW DATABASES;.
11. Set unimart as the default schema and verify it is visible in the Schemas panel.

```
[SQL]
SELECT VERSION();
USE unimart;
SELECT DATABASE();
SHOW TABLES;
```

5. Use the EER model from Guide 01

12. Open the saved .mwb model and compare table/column names with Java naming conventions.
13. Use snake_case in the database and camelCase in Java DTOs. Map explicitly when names differ.
14. Review indexes, foreign-key delete actions and generated DDL.
15. For the lab, let Flyway own schema creation. Use Workbench Forward Engineer only as a review tool unless instructed by the lecturer.
16. After the backend runs migrations, use Database > Reverse Engineer to compare the live schema with the model.

6. Safe data inspection and reset

```
[SQL]
-- Inspect without changing data.
SELECT id, university_email, role, created_at
FROM users
ORDER BY created_at DESC
LIMIT 20;

-- Never run destructive reset commands without a backup and lecturer approval.
-- DROP DATABASE unimart;
```

- Use explicit column lists instead of SELECT * in demonstrations.
- Limit result sets and never display password_hash or token data on a projector.
- Export schema/data before major migration exercises.
- Use a dedicated local database; never point a lab build at production data.

7. Backup and restore basics

```
[Terminal]
# Local command-line backup
mysqldump -u unimart_app -p --single-transaction \
  --routines --triggers unimart > unimart_backup.sql

# Restore into an empty development schema
mysql -u unimart_app -p unimart < unimart_backup.sql
```

8. Troubleshooting

Symptom	Check	Fix
Cannot connect to 127.0.0.1:3306	MySQL service status and port	Start service; verify port; check local firewall.
Access denied	Username, host and password	Use unimart_app@localhost; reset only through an admin account.
Unknown database unimart	Schema not created or wrong connection	Run CREATE DATABASE as admin; refresh Schemas.

Symptom	Check	Fix
Authentication plugin error	Old client/server mismatch	Update Workbench/Connector; use supported authentication.
Workbench EER feature issue with 8.4	Known version-line limitation	Use modelling features tested in class or model offline, then verify DDL manually.

Completion checklist

☐ MySQL server service starts successfully.

☐ Workbench connects using unimart_app rather than root.

☐ unimart schema uses utf8mb4.

☐ SQL query tab can run SELECT VERSION().

☐ The EER model can be opened and reviewed.

☐ A backup command has been tested with non-sensitive sample data.